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Axial Vibration Characteristics of Piecewise Large Space Truss Structures

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ABSTRACT

Large space truss structures (LSTS) are essential load-bearing components in many spacecraft. Owing to their large size, light-weight design, and high flexibility, they inherently exhibit very low natural frequencies and small damping ratios, making them susceptible to persistent vibrations induced by small disturbances. To address this issue, this study investigates a piecewise LSTS with tunable spring-damper joints, which both reflects the segmented assembly characteristics in practice and captures the interface discontinuities that significantly affect vibration characteristics. An analytical framework for the axial vibration of a piecewise elastic bar is developed using Hamilton's principle in a dimensionless form, whereas the boundary and transition conditions are obtained from the principle of minimum potential energy. The resulting characteristic equations are solved to determine the natural frequencies and discontinuous mode shapes, and the framework is further extended to forced vibration under harmonic and transient excitations. The analytical solutions are validated against finite element results, with relative errors in the first eight modes remaining below 1.3%. The results show that, compared with an equivalent uniform truss/bar, segmentation with compliant joints further reduces the natural frequencies because of local stiffness loss at the interfaces. Nevertheless, the displacement discontinuities at the joints localize elastic potential energy in the springs and enable its dissipation by the dampers, thereby reducing both steady-state and transient responses.

1 | Introduction

Trusses are widely used in bridges [1, 2], long-span roofs [3], lattice towers [4, 5], and space systems [6–9]. In space applications, simple truss assemblies provide the primary resistance to static and dynamic loads. Because they offer high stiffness and strength at very low mass, truss configurations are widely adopted in spacecraft primary structures [10]. Truss members and joints are inherently modular, which facilitates factory

prefabrication, transportation as small subassemblies, and rapid on-site bolted assembly [11].

Large space truss structures (LSTS) are therefore key load-bearing components of spacecraft [12–14]. These structures, which often extend from tens to hundreds of meters, are used to support sensitive payloads such as antennas and optical instruments, as illustrated in Figure 1. In practice, a space truss is usually divided into several segments, launched separately by

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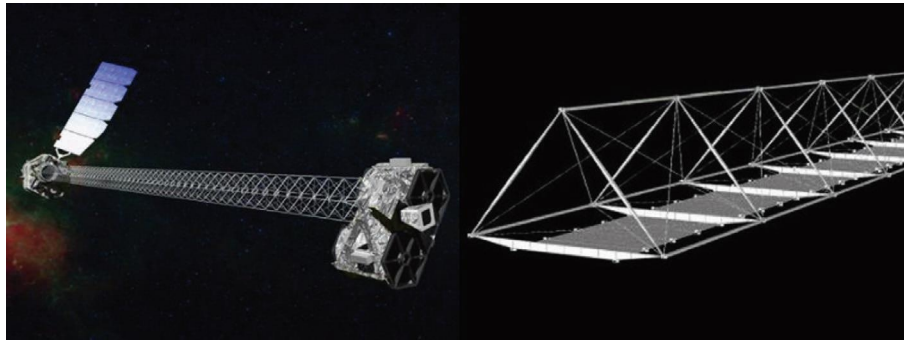


FIGURE 1 | The large space truss adapted from [15].

rockets, and finally assembled in orbit [16]. Manufacturing and assembly tolerances inevitably introduce gaps and clearances in the joints, which reduce the overall stiffness of the assembled truss [17].

In this study, an LSTS refers to a periodic structure composed of beams, rods, and cables that exhibits high flexibility and geometrical complexity. Its extremely low natural frequencies and small damping ratios make it susceptible to persistent vibrations induced by small disturbances, which can lead to structural fatigue [18, 19]. Consequently, effective vibration suppression of space trusses is crucial for extending service life and reducing control effort [20]. Various damping concepts have been proposed for LSTS. For example, Xu et al. [21] demonstrated the efficacy of viscoelastic dampers on mitigating vibrations, whereas Hu et al. [22] proposed a semiactive vibration control approach based on magnetically controlled joint stiffness and damping for flexible space solar panels, representing a class of joint-based hybrid damping strategies.

Recent reviews on space deployable mechanics highlight how advanced materials and intelligent actuation mechanisms expand the design space for lightweight, modular systems while introducing new dynamic uncertainties [23]. On-orbit assembly has emerged as a critical enabling technology for ultralarge space structures that exceed current launch-vehicle fairing limits. In this context, modular dynamic modeling techniques, such as the decoupling method proposed by Zhou et al. [24], enable scalable and efficient simulation of assembly processes. Hu et al. [25] further emphasized the indispensable role of accurate vibration modeling and robust control strategies in ensuring the operational reliability of such systems.

Parallel advances in structural dynamics and metamaterials show that periodic structures can tailor wave propagation and achieve targeted vibration attenuation [26, 27]. These mechanisms conceptually align with segmented space trusses, in which joints can serve not only as structural connections but also as engineered vibration-control elements. Experimental studies support this viewpoint: Siriguleng et al. [28] investigated modal coupling in deployable ring-truss antennas and showed that segmentation strongly influences dynamic response. From a modeling perspective, continuum approaches are particularly promising. Cao et al. [29] developed a coupled partial differential equation–ordinary differential equation continuum model that enables hub-only boundary control of flexible truss satellites, whereas Shi et al. [30] proposed simplified equivalent models that retain analytical tractability for vibration analysis of periodic ring-truss units.

A space truss can be viewed as a combination of multiple beams and often exhibits behavior similar to beam-like structures [31, 32]. Thanks to their simple geometry, beams possess well-understood mechanical behavior [33]. Gonella et al. [34] derived an equivalent elastic-rod model for the axial vibration of a one-dimensional beam-truss composed of planar truss elements. This suggests that, for axial dynamic analysis, a space truss can be simplified to a single bar, and a segmented space truss can be idealized as a set of connected bar segments.

Previous studies have recognised that joint flexibility and damping strongly influence the dynamic behavior of truss structures [35–37]. Belvin [38] developed an analytical joint model for truss structures using equivalent spring and damper elements, and validated it against experimental data, thereby establishing an early foundation for joint-level vibration analysis in large flexible spacecraft. More recently, Zhang et al. [39] analyzed multibeam systems with joints exhibiting stick-slip friction, using incremental harmonic balance to capture nonlinear joint effects on vibration characteristics. Zhang et al. also studied the dynamics of a two-segment deployable beam system and highlighted how segmentation fundamentally alters vibration responses [40]. Nieves et al. [41] considered transverse vibration of an axially compressed bar with dry friction at its ends, but restricted their formulation to transverse motion and neglected axial vibration. This simplification may be acceptable for small deflections and negligible axial preload but becomes questionable under appreciable axial compression.

Although these contributions underline the importance of joint dynamics and segmentation, they remain limited in scope: many focus on isolated joint behavior, consider beam-like structures rather than full trusses, or rely primarily on experimental observations without providing a unified analytical framework.

To address these limitations, the present work develops a theoretical model for piecewise elastic bars with tunable spring-damper joints. The principle of minimum potential energy (PMPE) and dimensionless analysis are used to derive natural frequencies, discontinuous mode shapes, and forced responses. The analytical framework is validated against finite element simulations and then applied to the axial vibration of a segmented space truss under both sinusoidal and rectangular pulse excitations. The results provide a theoretical basis for the passive vibration-control design of LSTS and related aerospace structures. The rest of this study is organised as follows. Section 1 formulates the physical model by idealizing the LSTS as a piecewise elastic bar connected by concentrated springs and

dampers, derives the boundary and transition conditions using PMPE, and obtains the dimensionless governing equations using Hamilton's principle. Section 2 presents the dimensionless analysis of piecewise-bar vibration characteristics, finite element verification, and axial vibration responses of space trusses under sinusoidal and pulse excitation. Section 3 summarizes the main findings and outlines implications for the design of LSTS.

2 | Model Development

On the basis of the piecewise bar model, an analytical framework is developed for axial vibration. The PMPE is employed to derive the boundary and transition conditions that govern the force–displacement relations at the fixed end, free end, and internal spring–damper joints. These conditions constitute the basis for the subsequent vibration analysis.

2.1 | Physical Model and Assumptions

The space truss considered here consists of periodically repeated bars with nearly identical geometry and material properties. Under small-amplitude motions and axial end excitation, the deformation field is dominated by axial extension and contraction of the bars. In this low-frequency regime, the internal axial degrees of freedom of one truss unit can be condensed into an equivalent axial stiffness [34]. Consequently, the whole truss can be represented as a piecewise uniform bar whose segments are connected by concentrated axial spring–damper elements, and the piecewise space truss can be simplified to segmented rods for analysis.

It should be noted that the present study is restricted to axial vibration. Under small-amplitude motion and axial end excitation, the dominant deformation of the periodic space truss is axial extension and contraction, so that the structural response can be reduced to an equivalent one-dimensional axial model. Accordingly, lateral bending and torsional deformations, as well as their coupling with axial motion, are not included in the present formulation. These effects are important for a more general three-dimensional dynamic analysis, but they require an equivalent beam/torsion model or a fully spatial model and are therefore beyond the scope of this study.

In addition, the present formulation is based on linear elastic and small-deformation assumptions. Structural nonlinearities, such as geometric nonlinearity induced by large axial preload, material nonlinearity, cable slackening, joint clearance, and contact nonlinearity, are not included. Therefore, the equivalent axial stiffness used in this study should be interpreted as the initial linear equivalent axial stiffness around the undeformed configuration. If a large axial force or prestress is present, the equivalent axial stiffness may become load-dependent, and an incremental tangent-stiffness formulation would be required. This nonlinear extension is beyond the scope of the present study.

Each segment is represented as a solid cylindrical bar, whereas the flexible joint between segments is modeled by a concentrated spring and a damper. Specifically, the axial stiffness of each bar segment is determined by its Young's modulus and

cross-sectional area, the joint flexibility is described by the stiffness of the concentrated spring, and the damping properties are represented by the coefficient of the concentrated damper.

Based on these assumptions, a piecewise bar model is established, as illustrated in Figure 2. The model consists of a segmented elastic bar undergoing axial vibration, with its left end fixed. Two internal hinge points are introduced along the x -axis. Here, $u_i s (i = 1, 2, 3)$ denotes the axial displacement field of the i -th bar segment. At the i -th internal interface, the left and right limiting displacements are denoted by u_i^- and u_i^+ , respectively. The piecewise bar is connected at two internal hinge points. Concentrated linear springs (stiffness k_1, k_2) and dampers (coefficients c_1, c_2) are embedded at hinge points $x = L_1$ and $x = L_2$. The bar is divided into three distinct segments along its length $L = L_3$.

It should be emphasized that the concentrated spring–damper at each internal joint does not represent a single physical device, but the lumped axial compliance and energy dissipation capacity of the segmented interface after condensation of its local joint members and connectors. In the present one-dimensional model, the relative displacement across the i -th internal interface, $u_i^+ - u_i^-$, is used to describe the local axial deformation of the joint. Accordingly, the axial restoring force and damping force transmitted by the interface are written as $F_{k,i} = k_i (u_i^+ - u_i^-)$ and $F_{c,i} = c_i (\dot{u}_i^+ - \dot{u}_i^-)$, respectively. Therefore, k_i characterizes the equivalent axial connection stiffness of the segmented joint, whereas c_i characterizes the equivalent viscous dissipation associated with the relative axial motion across that joint.

In practical implementation, the equivalent axial rigidity EA of each bar segment is obtained from the uniform space-truss model by a displacement-equivalence procedure: one end of the truss is fixed, a prescribed unit axial displacement is applied at the other end, and the resulting reaction force is used to calculate the equivalent axial stiffness and, hence, EA . The equivalent mass per unit length is determined by dividing the total mass of the space truss by its total length. For each segmentation interface, several local axial spring–damper branches exist in parallel in the three-dimensional truss model. Since these branches undergo the same interface axial displacement, the equivalent joint stiffness and damping adopted in the one-dimensional model are taken as the sums of the corresponding parallel contributions, that is, $k_i = \sum k_{ij}$ and $c_i = \sum c_{ij}$. In the verification example considered in this study, the two internal joints have the same geometric layout and connection form; therefore, $k_1 = k_2$ and $c_1 = c_2$ are used for simplicity. This treatment is a modeling simplification for a symmetric segmented truss, rather than a restriction of the proposed theoretical framework.

This piecewise bar model, together with the above parameter interpretation, provides the physical basis for the equivalent modeling and subsequent derivation of the governing equations of axial vibration. It should be further noted that the present equivalent bar model does not assume identical axial forces or stresses in all individual members of the space truss. In the actual three-dimensional structure, the longerons and diagonal cables may carry different local member forces because of their different geometries, orientations, and force-transfer roles. The purpose of the present simplification is instead to represent the resultant axial behavior of the whole truss unit under

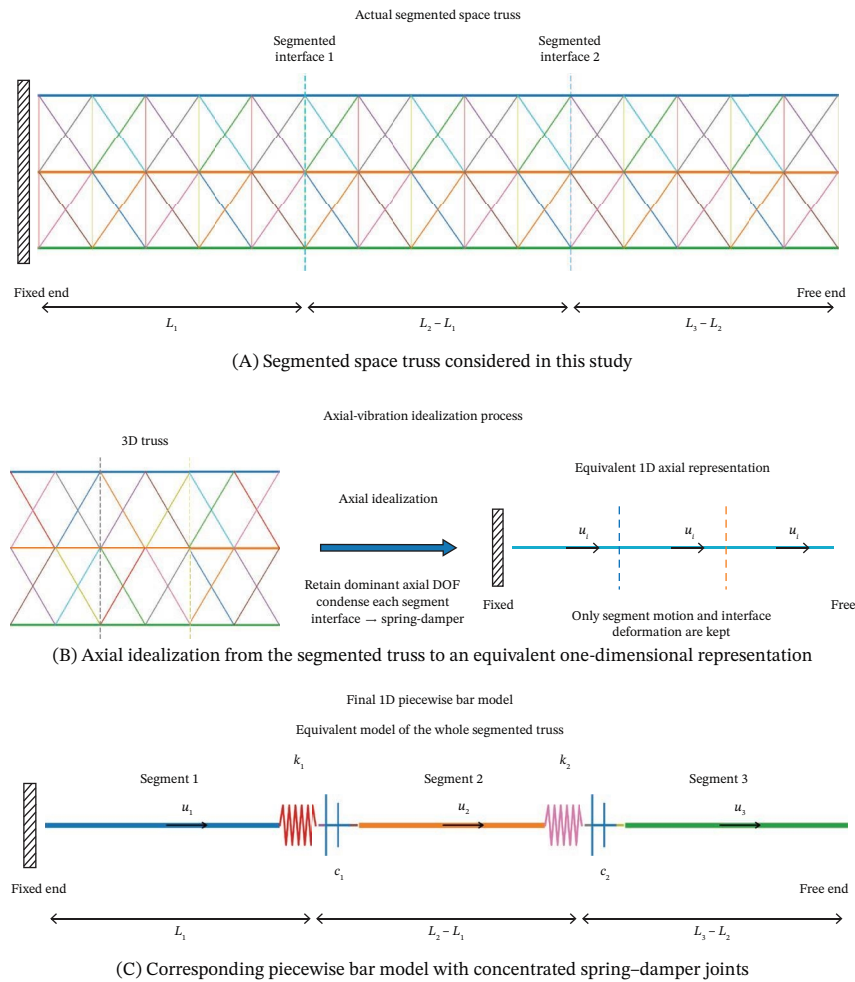


FIGURE 2 | Physical meaning of the proposed model. (A) Segmented space truss considered in this study. (B) Axial idealization from the segmented truss to an equivalent one-dimensional representation. (C) Corresponding piecewise bar model with concentrated spring–damper joints.

small-amplitude, low-frequency axial vibration. Therefore, the equivalent bar preserves the global axial force–displacement relation and the dominant low-frequency dynamic characteristics of the segmented truss, whereas member-level stress nonuniformity is absorbed into the equivalent parameters and is not resolved explicitly in the reduced-order model.

2.2 | Equivalent Modeling

Based on the physical interpretation of the joint parameters introduced in Section 2.1, the equivalent properties of the piecewise bar are determined herein using displacement-equivalence and energy-equivalence principles. In existing research, the vibration characteristics of space truss are often studied using methods such as energy equivalence criteria or displacement equivalence criteria, which treat space truss as equivalent to bars [36, 42, 43]. The displacement equivalence and energy-equivalence methods are used to model the piecewise space truss as piecewise bar elements in this study.

Using simulation software, one end of a uniform space truss, defined as a truss without segments, was fully fixed, and a prescribed axial displacement Δ was applied to the terminal

transverse section along the global axial direction. The total axial reaction force F was then extracted from the finite-element model. The equivalent axial stiffness and equivalent axial rigidity were calculated using the following formula.

$$k_{eq} = \frac{F}{\Delta}, \quad EA_{eq} = k_{eq}L = \frac{F}{\Delta}L. \quad (1)$$

Here, k_{eq} is the overall equivalent axial spring stiffness of the truss, EA_{eq} is the equivalent axial rigidity, L is the axial length of the truss in the loading direction, Δ is the imposed axial displacement, and F is the measured total axial reaction at the loaded end.

It should be noted that the prescribed unit displacement is only used as a numerical probing displacement in a linear elastic static analysis. Under the linear assumption adopted in this study, the reaction force is proportional to the imposed displacement, and the calculated equivalent axial stiffness is independent of the displacement magnitude. If large axial force or other nonlinear effects are considered, the equivalent stiffness would become load-dependent and should be evaluated using an incremental tangent-stiffness formulation.

At a segmentation plane L_p , the two interface cross-sections are modeled as rigid planes and only the axial DOF is retained. If N

axial springs are installed in parallel across L_i , each branch undergoes the same axial displacement Δ_{seg} . The total reaction force is

$$F = \sum_{j=1}^N k_j \Delta_{\text{seg}}. \quad (2)$$

Defining the equivalent axial spring by $F = k_{\text{eq}} \Delta_{\text{seg}}$ gives

$$k_{\text{seg}} = \sum_{j=1}^N k_j. \quad (3)$$

Here, k_j indicates the spring stiffness, and k_{seg} is the equivalent piecewise stiffness.

If N linear viscous dampers are installed **in parallel** across segmentation plane L_i , the equivalent joint damping is additive:

$$c_{\text{joint}} = \sum_{n=1}^N c_i. \quad (4)$$

Here, c_{joint} is the equivalent piecewise damping, and c_i is the damping of a spring damper in a segmented space truss.

2.3 | Boundary Conditions and Transition Conditions

In a piecewise bar system, the elastic potential energy stored in the concentrated spring can be written in the standard quadratic form used for equivalent joint modeling [38]:

$$V_{\text{spring},i} = \frac{1}{2} k_i (u_i^+ - u_i^-)^2. \quad (5)$$

Here, u_i^+ and u_i^- are the displacements of the right and left sides of the spring, and k_i is the stiffness of the i -th spring.

The elastic potential energy of an axially vibrating elastic bar segment is given by the classical strain-energy expression [44]:

$$V_{\text{bar},i} = \frac{1}{2} \int_{x_i}^{x_{i+1}} EA \left(\frac{\partial u_i}{\partial x} \right)^2 dx. \quad (6)$$

Here, E is the Young's modulus of the bar, and A is the cross-sectional area of the bar.

The PMPE is applied to determine the boundary conditions of the system. From Equations (5) and (6), the potential energy V of the system can be obtained as follows:

$$V = \int_0^{L_1} \frac{EA}{2} \left(\frac{\partial u_1}{\partial x} \right)^2 dx + \int_{L_1}^{L_2} \frac{EA}{2} \left(\frac{\partial u_2}{\partial x} \right)^2 dx + \int_{L_2}^{L_3} \frac{EA}{2} \left(\frac{\partial u_3}{\partial x} \right)^2 dx + \frac{k_1}{2} [u_2(L_1) - u_1(L_1)]^2 + \frac{k_2}{2} [u_3(L_2) - u_2(L_2)]^2 \quad (7)$$

Here, the springs and dampers are implicitly modeled as the concentrated ones. The bar is fixed at $x = 0$ and free at $x = L_3$.

The PMPE, that is, $\delta V = 0$ in conjunction with the integration by parts leads to following geometric and natural boundary conditions and transition conditions:

$$\begin{aligned} u_1(0) &= 0, EAu_{1x}(L_1) - k_1[u_2(L_1) - u_1(L_1)] = 0, \\ EAu_{2x}(L_1) - k_1[u_2(L_1) - u_1(L_1)] &= 0, \\ EAu_{2x}(L_2) - k_2[u_3(L_2) - u_2(L_2)] &= 0, \\ EAu_{3x}(L_2) - k_2[u_3(L_2) - u_2(L_2)] &= 0, u_{3x}(L_3) = 0 \end{aligned} \quad (8)$$

Here, $u_{ix} = \partial u_i / \partial x$; we have one boundary condition at $x = 0$ and $x = L_3$; $u_1(0) = 0$ means no displacement at the fixed end and $u_{3x}(L_3) = 0$ means no stress at the free end. This boundary condition is consistent with the boundary condition derived by Zhang et al. [45] using the principle of virtual work.

In contrast, two additional boundary conditions emerge at $x = L_1$ and $x = L_2$ where springs and dampers are located. Strictly speaking, they are not boundary conditions because they do not occur at the boundary. They are termed "transition conditions" [46], which are $EAu_{1x}(L_1) = EAu_{2x}(L_1)$ and $EAu_{2x}(L_2) = EAu_{3x}(L_2)$ as given by Equation (8), which embody Newton's third law at $x = L_1$ and $x = L_2$. The four transition conditions at $x = L_1$ and $x = L_2$ mathematically ensure the continuity of the elastic potential under the assumption of uniform axial rigidity EA . In general, the elastic potential is discontinuous. In contrast, the displacements at $x = L_1$ and $x = L_2$ are always discontinuous.

For a general piecewise bar with n segments, the displacement field in the i -th segment is denoted by $u_i(x, t)$, where $x \in (L_{i-1}, L_i)$ and $i = 1, 2, \dots, n$. In the following nondimensionalization, $U_i = u_i / L$ is used for the i -th segment. To nondimensionalize Equation (8), the following dimensionless variables are introduced:

$$\xi = \frac{x}{L}, \quad \xi_1 = \frac{L_1}{L}, \quad \xi_2 = \frac{L_2}{L}, \quad U = \frac{u}{L}, \quad \tau = \sqrt{\frac{E}{\rho}} \frac{t}{L}, \quad K_i = \frac{k_i L}{EA}, \quad D = \sqrt{\frac{\rho}{E}} \frac{cL}{m}, \quad D_c = \sqrt{\frac{\rho}{E}} \frac{c_1}{m} \quad (9)$$

Here, $L = L_3$ is the bar length, and ξ_1 and ξ_2 are the dimensionless locations of the concentrated springs and dampers.

Equation (8) is now nondimensionalized as

$$\begin{aligned} \frac{\partial U_1(0, \tau)}{\partial \xi} &= 0, \quad \frac{\partial U_1(\xi_1, \tau)}{\partial \xi} - K_1[U_2(\xi_1, \tau) - U_1(\xi_1, \tau)] = 0, \\ \frac{\partial U_2(\xi_1, \tau)}{\partial \xi} - K_1[U_2(\xi_1, \tau) - U_1(\xi_1, \tau)] &= 0, \\ \frac{\partial U_2(\xi_2, \tau)}{\partial \xi} - K_2[U_3(\xi_2, \tau) - U_2(\xi_2, \tau)] &= 0, \\ \frac{\partial U_3(\xi_2, \tau)}{\partial \xi} - K_2[U_3(\xi_2, \tau) - U_2(\xi_2, \tau)] &= 0, \quad \frac{\partial U_3(\xi_3, \tau)}{\partial \xi} = 0. \end{aligned} \quad (10)$$

2.4 | Governing Equations

Applying Hamilton's principle to the longitudinal vibration of an elastic bar yields the classical governing equation [44]:

$$m_i \frac{\partial^2 u_i}{\partial t^2} + E_i A_i \frac{\partial^2 u_i}{\partial x^2} = 0, \quad x \in (L_{i-1}, L_i), \quad i = 1, 2, \dots, n. \quad (11)$$

For the symmetric three-segment model considered in this study, the equivalent properties of all segments are taken to be identical, that is, $m_i = m$ and $E_i A_i = EA$, so the above system reduces to the compact form used hereafter. Here, $m = \rho A$ (ρ : density) is the mass per unit length. Equation (11) is nondimensionalized as the following:

$$\frac{\partial^2 U}{\partial \tau^2} + \frac{\partial^2 U}{\partial \xi^2} = 0. \quad (12)$$

The solution form of $U(\xi, \tau) = \psi(\xi)T(\tau)$ is assumed and substituted into Equation (12), which yields the following:

$$\frac{\ddot{T}}{T} = -\frac{\ddot{\psi}}{\psi} = -\omega^2. \quad (13)$$

Here, $\psi(\xi)$ denotes the dimensionless spatial mode shape (eigenfunction) obtained after separation of variables. Here, $\ddot{T} = \partial^2 T / \partial \tau^2$ and $\ddot{\psi} = \partial^2 \psi / \partial \xi^2$; ω is the natural frequencies to be determined by the boundary and transition conditions. The solution form of ψ can be obtained from the above Equation (13) as follows:

$$\psi(\xi) = \begin{cases} \psi^1(\xi) = A_1 \cos(\omega \xi) + B_1 \sin(\omega \xi) & 0 \leq \xi \leq \xi_1 \\ \psi^2(\xi) = A_2 \cos(\omega \xi) + B_2 \sin(\omega \xi) & \xi_1 \leq \xi \leq \xi_2 \\ \psi^3(\xi) = A_3 \cos(\omega \xi) + B_3 \sin(\omega \xi) & \xi_2 \leq \xi \leq \xi_3 \end{cases} \quad (14)$$

There are a total of six unknowns: A_i and B_i ($i = 1, 2, 3$) in Equation (10), and the six boundaries and transition conditions are used to find the natural frequencies and mode shapes. For simplicity, we set $K = K_1 = K_2$ and $\xi_1 = 1/3, \xi_2 = 2/3$. Substitute Eq. into Equation (14) to obtain six sets of equations, and write the six sets of equations in matrix form $MX = 0$, where $A_1 = 0, X = [B_1, A_2, B_2, A_3, B_3]^T$.

$$M(\omega) = \begin{pmatrix} \omega \cos(\omega \xi_1) + K_1 \sin(\omega \xi_1) & -K_1 \cos(\omega \xi_1) & -K_1 \sin(\omega \xi_1) & 0 & 0 \\ K_1 \sin(\omega \xi_1) & -\omega \sin(\omega \xi_1) - K_1 \cos(\omega \xi_1) & \omega \cos(\omega \xi_1) - K_1 \sin(\omega \xi_1) & 0 & 0 \\ 0 & -\omega \sin(\omega \xi_2) - K_2 \cos(\omega \xi_2) & \omega \cos(\omega \xi_2) - K_2 \sin(\omega \xi_2) & -K_2 \cos(\omega \xi_2) & -K_2 \sin(\omega \xi_2) \\ 0 & K_2 \cos(\omega \xi_2) & K_2 \sin(\omega \xi_2) & -\omega \sin(\omega \xi_2) - K_2 \cos(\omega \xi_2) & \omega \cos(\omega \xi_2) - K_2 \sin(\omega \xi_2) \\ 0 & 0 & 0 & -\sin \omega & \cos \omega \end{pmatrix}. \quad (15)$$

Make the determinant of the coefficient matrix zero.

$$\det(M(\omega)) = 0. \quad (16)$$

The solution to Equation (16) is the dimensionless natural frequency of the piecewise bar. For each natural frequency, substitute the matrix equation to solve for the corresponding nonzero

vector X_i , that is, the modal coefficient, and then obtain the mode shape of that order from Equation (14).

The derivations of boundary and transition conditions, natural frequencies, and mode shapes presented above are confined to concentrated spring systems, in accordance with standard analytical frameworks. Note that damping effects are excluded per the velocity-independent nature of static boundary conditions.

2.5 | Forced Vibration Analysis

To capture realistic operating conditions, the formulation is extended to include distributed and concentrated damping as well as external support excitation. The resulting governing equations are expressed in dimensionless form and represented in modal coordinates. Because closed-form solutions are generally not available for such forced vibration problems, the responses are computed numerically using the Galerkin method [47].

The effect of damping is incorporated into the forced-vibration formulation. For a general piecewise bar with n segments, the governing equations can be written in a compact whole-domain form as follows:

$$m \frac{\partial^2 u_{in}}{\partial t^2} + c \frac{\partial u_{in}}{\partial t} - EA \frac{\partial^2 u_{in}}{\partial x^2} - k[(u_2 - u_1)\delta(x - L_1) + (u_3 - u_2)\delta(x - L_2)] + c_1 \operatorname{sgn}\left[\left(\frac{\partial u_2}{\partial t} - \frac{\partial u_1}{\partial t}\right)\delta(x - L_1)\right] + c_2 \operatorname{sgn}\left[\left(\frac{\partial u_3}{\partial t} - \frac{\partial u_2}{\partial t}\right)\delta(x - L_2)\right] = 0 \quad (17)$$

Equation (17) is written in a compact whole-domain form. Equivalently, it represents the segment-wise forced-vibration equations for $u_i(i, t)$, coupled by the spring-damper transition conditions at the internal interfaces. Unless otherwise stated, the subsequent derivation is presented for the symmetric three-segment verification model, for which identical segment properties are assumed to simplify notation. Here, $u_{in,i}(x, t)$ denotes the axial displacement of the i -th segment in the inertial frame. For the i -th segment, the axial displacement in the inertial frame is written as $u_{in,i}(x, t) = u_0(t) + u_i(x, t)$, where $u_0(t)$ is the support excitation and $u_i(x, t)$ is the axial displacement of the i -th segment relative to the fixed base. Here, c and c_i are the damping coefficients of the bar segment and the i -th concentrated damper, respectively. For the symmetric three-segment model consid-

ered in this study, $c_1 = c_2$ is assumed for simplicity. The function $\operatorname{sgn}(\cdot)$ denotes the sign function and is introduced to preserve the positive dissipation effect of the concentrated dampers.

$$\operatorname{sgn}(y) = \begin{cases} -1, & y < 0 \\ 0, & y = 0 \\ 1, & y > 0 \end{cases} \quad (18)$$

The $u_{in}(x, t) = u_0(t) + u(x, t)$, $u_0(t)$ is assumed the support excitation and $u(x, t)$, as defined before, is the bar displacement relative to fixed end. Here, c and c_1/c_2 are the damping coefficients of the bar and concentrated damper, respectively. It is worth mentioning that c and c_1/c_2 are with different units. For simplicity, we set $c_1 = c_2$. Moreover, sgn is the sign function, which is to ensure the effect of positive damping of the concentrated dampers. With the dimensionless scheme as given in Equation (9), Equation (17) is now nondimensionalized as

$$\frac{\partial^2 U}{\partial \tau^2} + D \frac{\partial U}{\partial \tau} - \frac{\partial^2 U}{\partial \xi^2} - K[(U_2 - U_1)\delta(\xi - \xi_1) + (U_3 - U_2)\delta(\xi - \xi_2)] + D_c \text{sgn}[(\dot{U}_2 - \dot{U}_1)\delta(\xi - \xi_1) + (\dot{U}_3 - \dot{U}_2)\delta(\xi - \xi_2)] = -\frac{\partial^2 U_0}{\partial \tau^2} - D \frac{\partial U_0}{\partial \tau} \quad (19)$$

Here, D and D_c , as defined in Equation (9), are the dimensionless damping of the bar and concentrated dampers; $-(\partial^2 U_0)/(\partial \tau^2)$ is the inertial force due to the support excitation and $-D(\partial U_0)/(\partial \tau)$ is due to the bar damping. In our computation, we assume $-(\partial^2 U_0)/(\partial \tau^2)$ is a dominant one and $-D(\partial U_0)/(\partial \tau)$ is rather small for simplicity reason. Now we assume

$$U(\xi, \tau) = \sum_{j=1}^N a_j(\tau) \psi_j(\xi). \quad (20)$$

Here, N denotes the modal truncation order, that is, the number of retained modes in the Galerkin expansion, and is not related to the number of physical segments of the piecewise bar. The number of segments is denoted separately by n in the governing equations for the general piecewise bar. The value of N is determined from a convergence test. In the present study, $N = 5$ is adopted because the relative error of the maximum axial displacement at the free end, with respect to the reference solution obtained using $N = 10$, is already below 0.05%.

In this section, we present a dimensionless formulation and numerical results for the natural frequencies and discontinuous mode shapes of the piecewise bar system. It also formulates governing equations for forced vibration, where the free vibration characteristics are derived analytically and the forced responses are solved numerically in modal space.

3 | Results and Discussions

3.1 | Convergence Test

Before starting the dynamic analysis of the LSTS, a convergence test is conducted to determine the appropriate modal truncation order N for reliable numerical results. We assess the effect of modal truncation using the **tip displacement** $u(L, t)$ of the piecewise bar as the only observable. The idea is to keep the physical model, damping, and loading unchanged, increase the number of retained modes N , and check whether $u(L, t)$ stabilizes as N grows within prescribed tolerances.

To clarify the convergent behavior of the response, the values of the maximum axial displacement for each N are shown in

Table 1, where the errors are computed using the value when $N = 10$ as the reference value. As shown in this table, the relative error decreases rapidly with the increase of N . Therefore, the truncated Number 5 is used in the further analysis of this study since the relative error is less than 0.05%.

3.2 | Dimensionless Results

In order to analyze the relationship between the natural frequency of the piecewise bar and the spring stiffness K , Equation (16) is solved in MATLAB for a range of K values. It is clear from Figure 3 that as the dimensionless concentrated spring stiffness K increases, the natural frequencies ω_i ($i = 1, 2, 3, 4$) increases monotonically to $(2i - 1)\pi/2$, which is the natural frequencies of a uniform bar [44].

For $K > 0$, all natural frequencies of the piecewise bar system are consistently lower than their uniform bar counterparts.

TABLE 1 | Effect of the truncated number on the maximum axial displacement.

Number of retained modes (N)	Maximum axial displacement of the free end (mm)	Relative error (%)
1	145.1342	1.3665
2	146.8009	0.2338
3	146.6877	0.3107
4	147.3975	0.1716
5	147.2136	0.0466
6	147.2206	0.0513
7	147.1750	0.0203
8	147.1458	0.0005
9	147.1475	0.0017
10	147.1450	0.0000

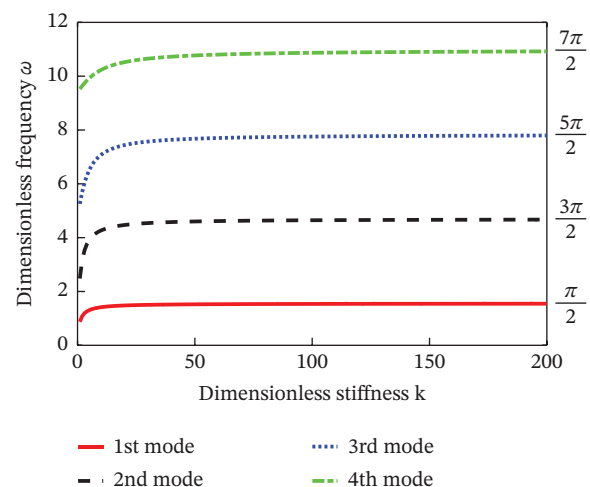


FIGURE 3 | The first four dimensionless natural frequencies of the piecewise bar system as a function of spring stiffness.

This observation indicates that the introduction of concentrated springs at ξ_1 and ξ_2 reduces the overall effective axial stiffness of the structural system compared with the uniform bar. This comparison is relative to the equivalent uniform bar used as the reference model.

The first four mode shapes of the piecewise bar are obtained by solving Equation (14) for $k = 50$ and $k = 10,000$. For $k \geq 10,000$, the spring in the piecewise bar is approximated as a uniform bar. The displacement discontinuity is more clearly seen in the mode shapes of the hinged bar as presented in Figure 4. As discussed in the boundary and transition conditions of Equation (8), which ensure the Newton's third law and thus cause the discontinuity of the displacements at ξ_1 and ξ_2 , this displacement discontinuity, or say, the displacement discontinuity are responsible for the stiffness reduction of the hinged bar. It should be noted that this trend is defined relative to the equivalent uniform bar used as the reference model. It does not contradict the fact that LSTS themselves are intrinsically low-frequency flexible structures; rather, it indicates that segmentation further decreases the natural frequencies

because of the local stiffness loss introduced by the compliant interfaces.

For a piecewise point at $x = \xi_i$, the condition:

$$\left. \frac{\partial U}{\partial \xi} \right|_{\xi_i^-} = K_i [U(\xi_i^+) - U(\xi_i^-)]. \quad (21)$$

The term $\partial U / \partial \xi|_{\xi_i^-}$ represents the axial force transmitted from the left bar segment to the spring.

$$\frac{F_i}{EA} = K_i [U(\xi_i^+) - U(\xi_i^-)]. \quad (22)$$

It follows from Equation (22) that this force equals the reaction force in the spring:

Consequently, traction continuity at the piecewise point requires the following:

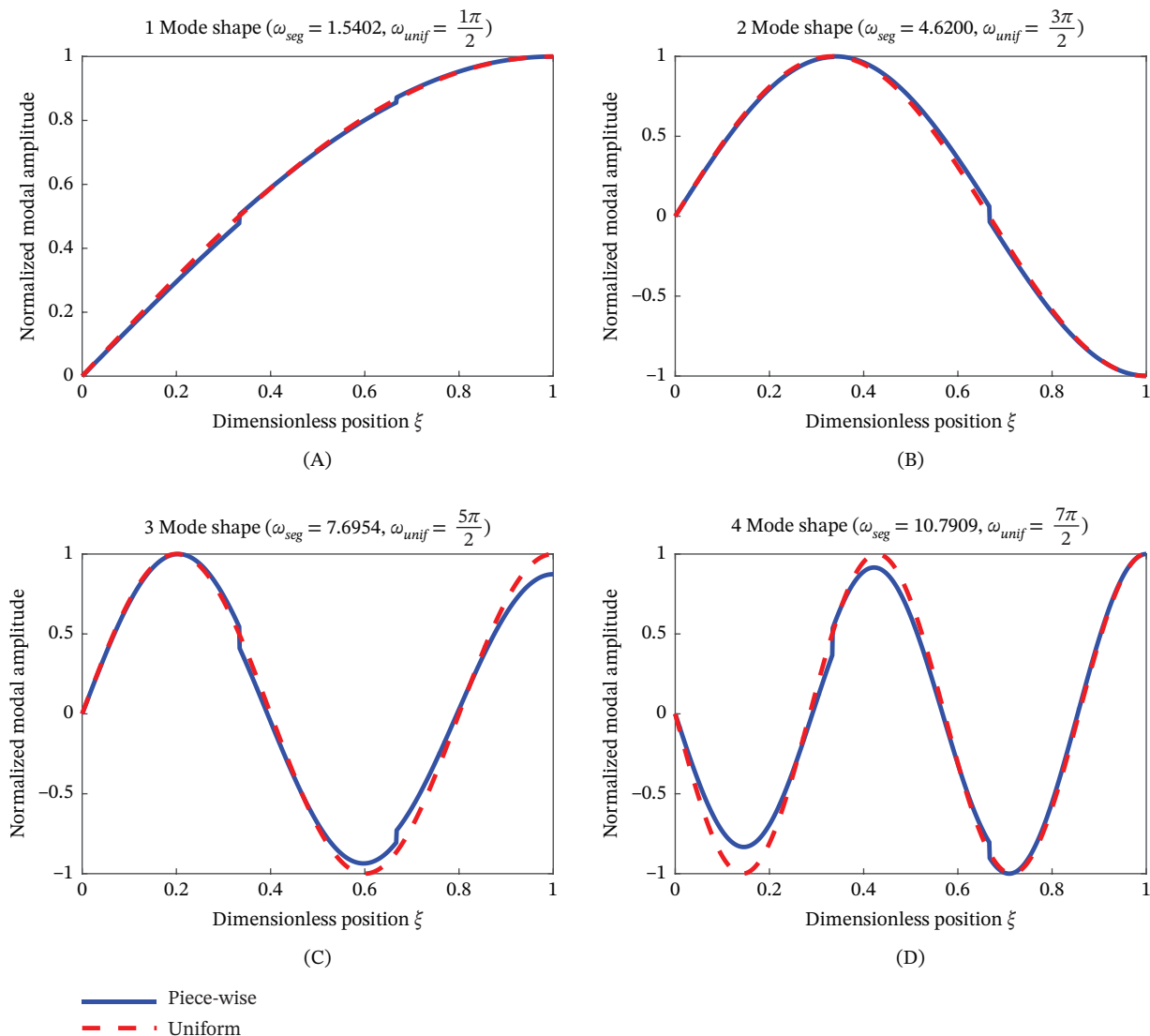


FIGURE 4 | First four dimensionless mode shapes of the piecewise bar system: (A) First mode, (B) second mode, (C) third mode, and (D) fourth mode.

$$\left. \frac{\partial U}{\partial \xi} \right|_{\xi_i^-} = \left. \frac{\partial U}{\partial \xi} \right|_{\xi_i^+}. \quad (23)$$

Ensuring axial force continuity across the discontinuity, the displacement discontinuity phenomenon occurring at the concentrated spring indicates that the concentrated spring has stored a large amount of energy instantaneously.

To comprehensively evaluate the vibration characteristics of the piecewise truss, two types of external excitations are applied. Sinusoidal excitation represents a sustained harmonic input, allowing analysis of the system's steady-state frequency response and resonance behavior. This reflects periodic disturbances commonly encountered in aerospace applications. In contrast, rectangular pulse excitation represents a sudden disturbance, such as a shock or impact, and is used to assess the transient response and damping effectiveness. Considering both a sustained harmonic input and a finite-duration pulse allows us to characterize the steady-state frequency response and the transient decay, respectively. Note that under harmonic excitation the response also contains an initial transient that decays with

damping, whereas a single pulse produces only transient behavior after the excitation ends. Using Equation (20), the dynamic response is computed with the modal expansion truncated to the first five modes.

In Figure 5, we compare the dynamic responses of the piecewise bar linear system and a uniform bar at $\xi = 0.3$, $\xi = 0.5$, $\xi = 0.7$, and $\xi = 1$. Here, $\xi = 0.3$ is the location just before the concentrated springs and dampers location of $\xi = 1/3$; $\xi = 0.5$ is the location between the concentrated springs and dampers location of $\xi = 1/3$ and $\xi = 2/3$; $\xi = 0.7$ is the location just behind the concentrated spring and damper location of $\xi = 2/3$; $\xi = 1$ is the free end of the bar. The sinusoidal excitation of $-(\partial^2 U_0) / (\partial \tau^2) = \sin(4\tau)$, the action direction is axial, and it is a global excitation. The “global excitation” in this study refers to the spatially uniform base acceleration along the bar axis, which generates equivalent inertial body forces on all mass elements throughout the truss. $D = 0.3, D_c = 10, K = 150$, and $N = 5$ are set. At $\xi = 0.3$, $\xi = 0.5$, $\xi = 0.7$, and $\xi = 1$, the responses of the piecewise bar linear system are significantly less than those of a uniform bar. However, at $\xi = 0.7$, the response of the piecewise linear system is a little bit larger. The presence of concentrated

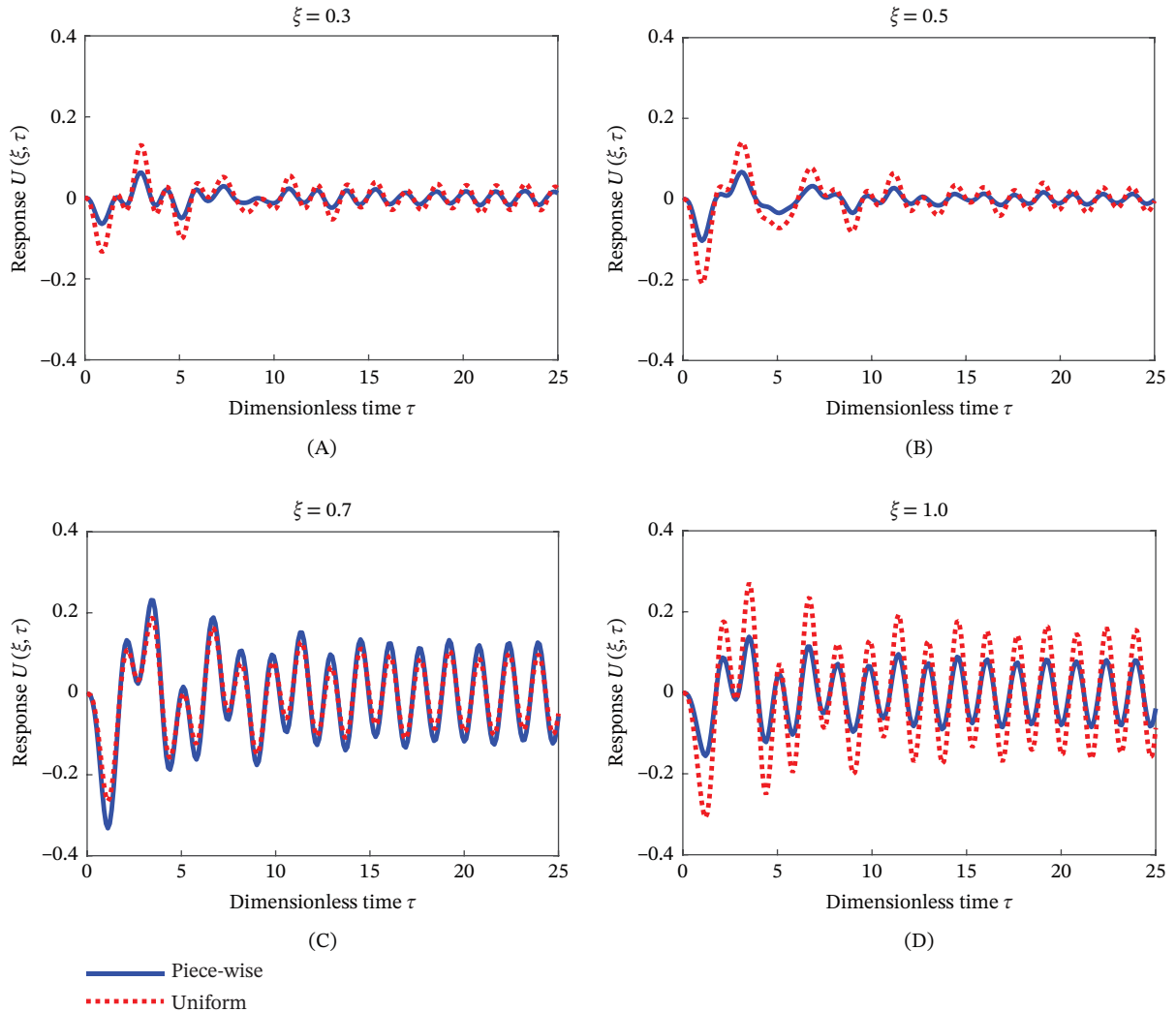


FIGURE 5 | Dynamic responses $U(\xi, \tau)$ of the piecewise bar and the uniform bar under a sinusoidal excitation. Panels correspond to different measurement positions: (A) $\xi = 0.3$, (B) $\xi = 0.5$, (C) $\xi = 0.7$, and (D) $\xi = 1$.

spring leads to the discontinuity of displacement and velocity, which in essence redistribute the energy along the bar: Some energy is stored in the concentrated springs and the concentrated dampers further dissipate more energy. As a whole, the energy inside the piecewise bar is less than that of a uniform bar. However, here, we need to emphasize this the piecewise linear system as a whole is actually a nonlinear system. It is possible that at some places the response of the piecewise linear system is large.

In Figure 6, the excitation is replaced as a rectangular pulse, which is defined as follows:

$$-\frac{\partial^2 U_0}{\partial \tau^2} = \begin{cases} 1, & 0 \leq \tau \leq 1 \\ 0, & \tau > 1 \end{cases}. \quad (24)$$

It is clear for this rectangular pulse excitation, the responses of the piecewise system decay much faster at all locations.

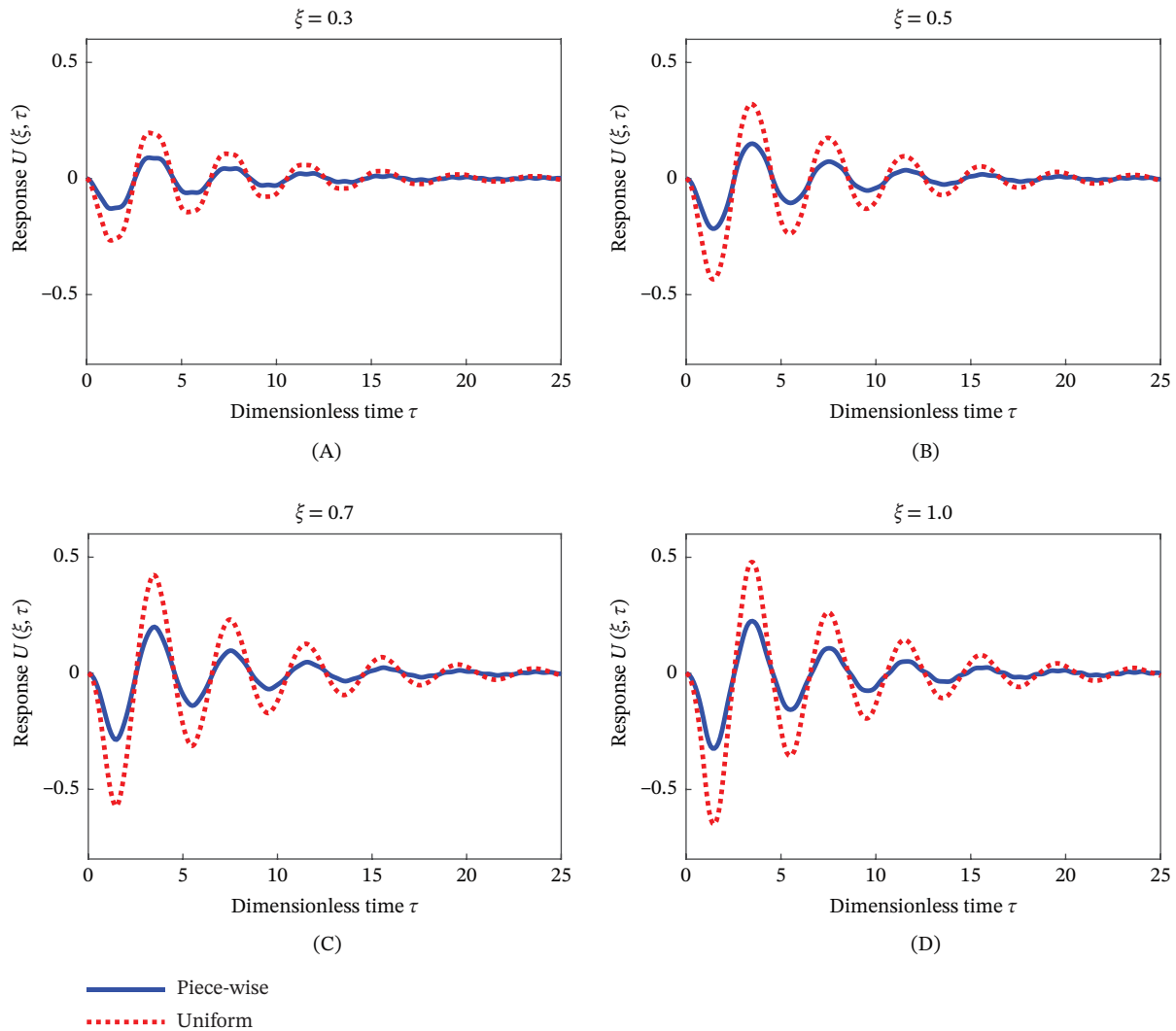


FIGURE 6 | Comparison of dynamic responses of the piecewise bar system and a uniform bar at (A) $\xi = 0.3$, (B) $\xi = 0.5$, (C) $\xi = 0.7$, and (D) $\xi = 1$ under a rectangular pulse excitation.

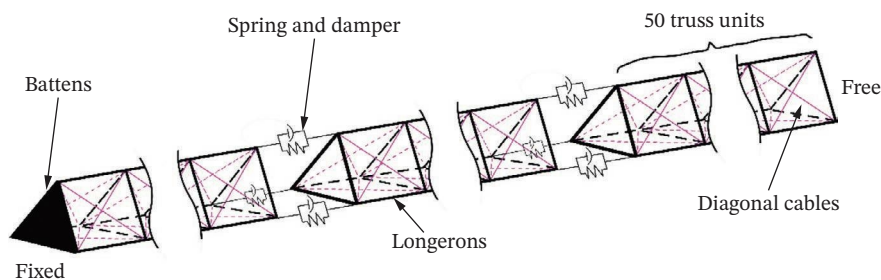


FIGURE 7 | Piecewise space truss.

This displacement discontinuity results in four major effects: The first is the changes of natural frequencies as seen in Figure 3; the second is the changes of mode shapes as seen in Figure 4; the third is that some energy is stored in the concentrated springs, as shown in Equation (7). and furthermore, the concentrated springs act as impedance that impacts the transmission and reflection of the longitudinal waves propagation from one section to another; as Figures 5 and 6, this displacement discontinuity also leads to the discontinuity of velocity and a concentrated damper can thus dissipate the energy.

3.3 | FEA of Space Piecewise Space Truss

To verify the engineering applicability of the above dimensionless analysis results, we now extend this theoretical framework to investigate the axial vibration characteristics of LSTS, a full-scale FE piecewise space truss model was constructed, with a total length up to 300m, significantly exceeding conventional aerospace structures, qualifying as “large-scale.” The LSTS is modeled as a cantilever. The root end at $x = 0$ is fixed, following a common idealization for spacecraft-attached truss structures [48]. The three longerons root nodes are fully clamped, with all translational and rotational degrees of freedom fixed, so the

axial displacement at the root satisfies $u(0, t) = 0$, on the other end is free. In the rest of this study, the cantilever piecewise space truss shown in Figure 7 is taken as the research object, for which the geometrical and material parameters of the space truss are summarized in Table 2.

The cross-sections of the battens and the longerons are hollow circular. The cross-section of the diagonal cable is a solid circle. The parameters of each spring are the same, and the parameters of each damper are also the same. Each piecewise space truss has 50 truss units that are rigidly connected to each other. The simplified form of the actual piecewise space truss consists of 150 truss units. A truss unit consists of three longerons, six battens, and six diagonal cables. By adjusting the stiffness coefficient k_i of the spring and the damping coefficient c_i of the damper, the degree of segmentation of the space truss can be changed.

Equate the segmented space truss to a piecewise bar physical model. The parameters of the physical model are shown in Table 3:

The equivalent properties listed in Table 3 were determined from the three-dimensional segmented space-truss model by using displacement-equivalence and interface condensation. First, the equivalent axial rigidity EA was identified from a uniform space-truss model without segmentation. One end of the truss was fixed, and a prescribed unit axial displacement was

TABLE 2 | Geometrical and material parameters of the LSTS.

Parameters	Values
Young's modulus of the battens	90 GPa
Density of the battens	1800 kg/m ³
Length of the battens	1 m
Poisson's ratio of the battens	0.3
Inter diameter of the battens	0.038 m
Outer diameter of the battens	0.04 m
Young's modulus of the longerons	210 GPa
Density of the longerons	1600 kg/m ³
Length of the longerons	2 m
Poisson's ratio of the longerons	0.3
Inter diameter of the longerons	0.038 m
Outer diameter of the longerons	0.04 m
Young's modulus of the diagonal cable	210 GPa
Density of the diagonal cable	1600 kg/m ³
Length of the diagonal cable	2.236 m
Poisson's ratio of the diagonal cable	0.3
Diameter of the diagonal cable	0.002 m
Total length of space truss	300 m
Position of the first spring and damper	100 m
Position of the second spring and damper	200 m
Stiffness of spring	3.33×10^7 N/m
Damping coefficient of the damper	3.33×10^7 N·s/m

TABLE 3 | Equivalent properties of the piecewise bar obtained from the space-truss model.

Parameters	Values
Total length of piecewise bar	300 m
Position of the first spring and damper	100 m
Position of the second spring and damper	200 m
Axial stiffness of piecewise bars (EA)	920919205 N
Mass per unit length	55.4286 kg/m
Stiffness of spring	10^8 N/m
Damping coefficient of the damper	10^8 N·s/m

TABLE 4 | Comparison of natural frequencies between the piecewise bar system and finite element method.

Mode order	FEA (Hz)	Analytical solution(Hz)	Error (%)
1	3.2741	3.2966	0.6900
2	9.8145	9.8867	0.7400
3	16.2900	16.4569	1.0200
4	23.0460	23.1253	0.3400
5	29.46300	29.6605	0.6700
6	35.7100	36.1754	1.3000
7	43.0310	43.0486	0.0400
8	49.1670	49.4368	0.5500

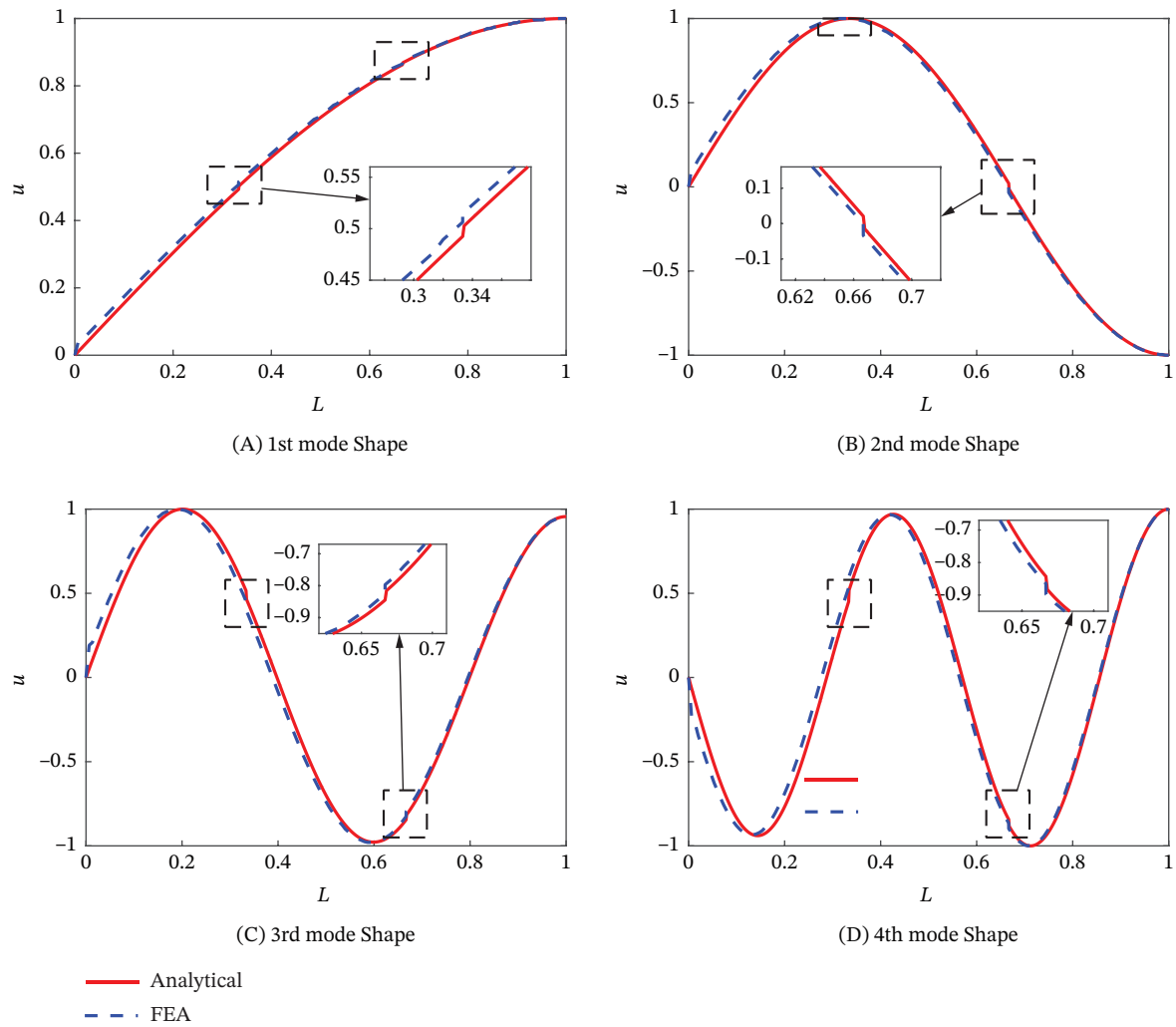


FIGURE 8 | First four mode shapes of the piecewise space truss obtained from finite element method. (A) First mode shape. (B) Second mode shape. (C) Third mode shape. (D) Fourth mode shape.

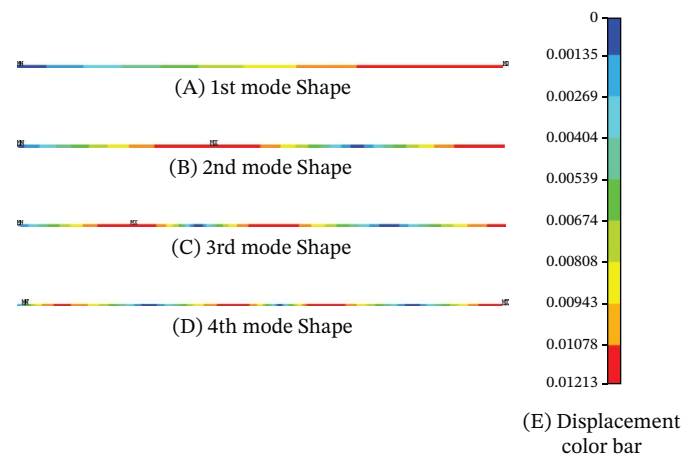


FIGURE 9 | Mode shapes of piecewise space truss. (A) First mode shape. (B) Second mode shape. (C) Third mode shape. (D) Fourth mode shape. (E) Displacement color bar.

applied at the other end. The resulting axial reaction force was then used to calculate the equivalent axial stiffness, from which the equivalent axial rigidity EA was obtained.

The quantity listed in Table 3 as mass per unit length was determined by dividing the total mass of the space truss by its total length.

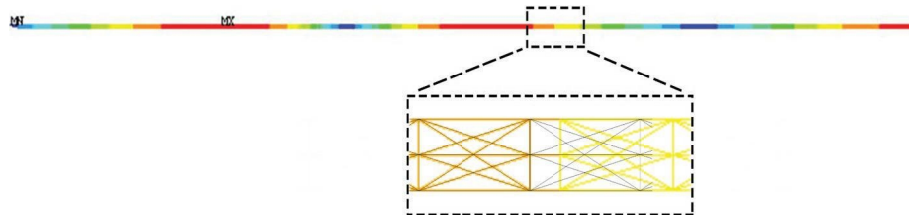


FIGURE 10 | Enlarged view of the displacement discontinuities for the third mode shape of the piecewise space truss.

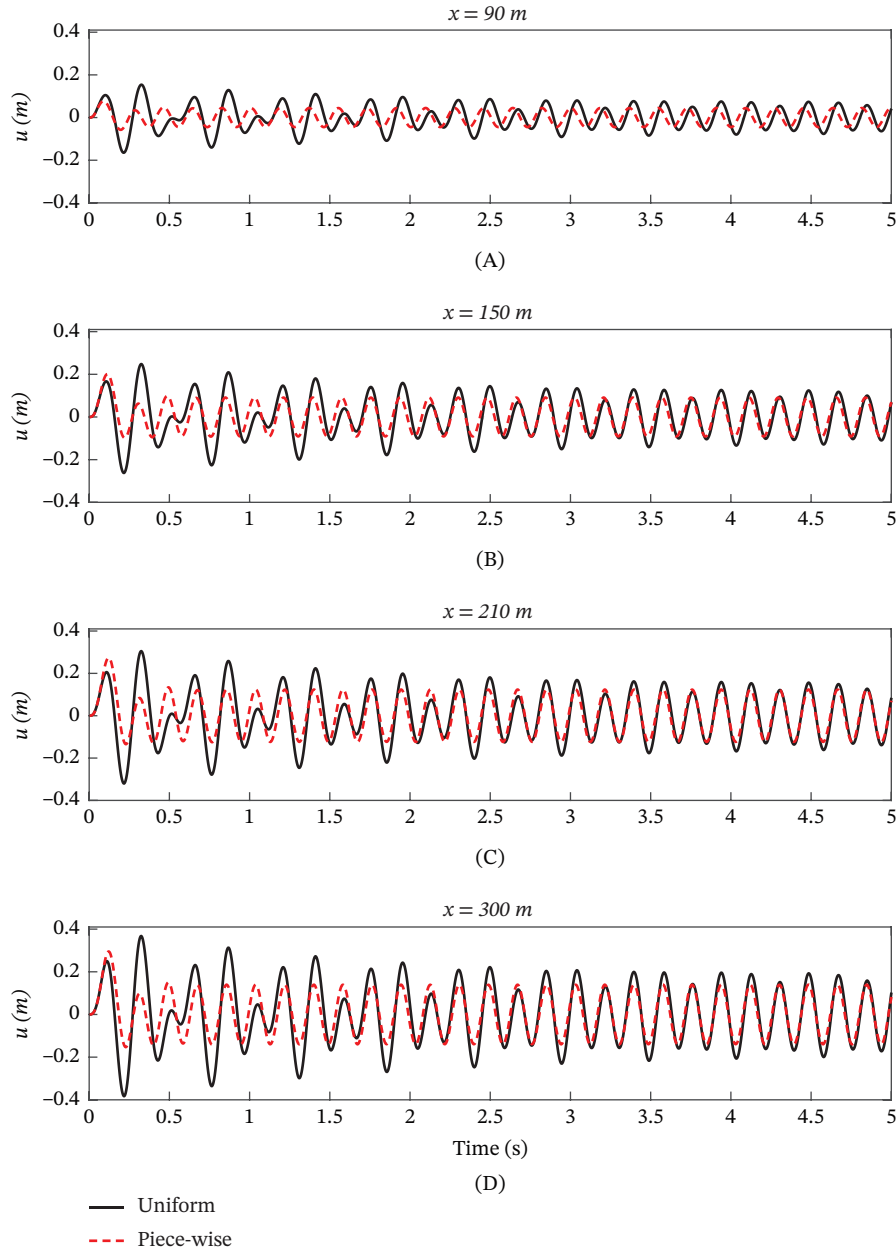


FIGURE 11 | Finite element results of axial dynamic responses of the piecewise space truss and uniform truss. (A) 90, (B) 150, (C) 210, and (D) 300 m.

At each segmentation interface of the three-dimensional space-truss model, three axial spring-damper branches are connected in parallel. Because these branches experience the same interface axial displacement, the equivalent spring stiffness and damping coefficient in the reduced one-dimensional model are obtained by summing the parallel contributions.

Therefore, the equivalent interface stiffness and damping used in Table 3 are three times those of an individual branch in the space-truss model. In the present verification model, the two internal interfaces have the same geometric layout and connection form, and thus identical equivalent parameters are adopted, that is, $k_1 = k_2$ and $c_1 = c_2$.

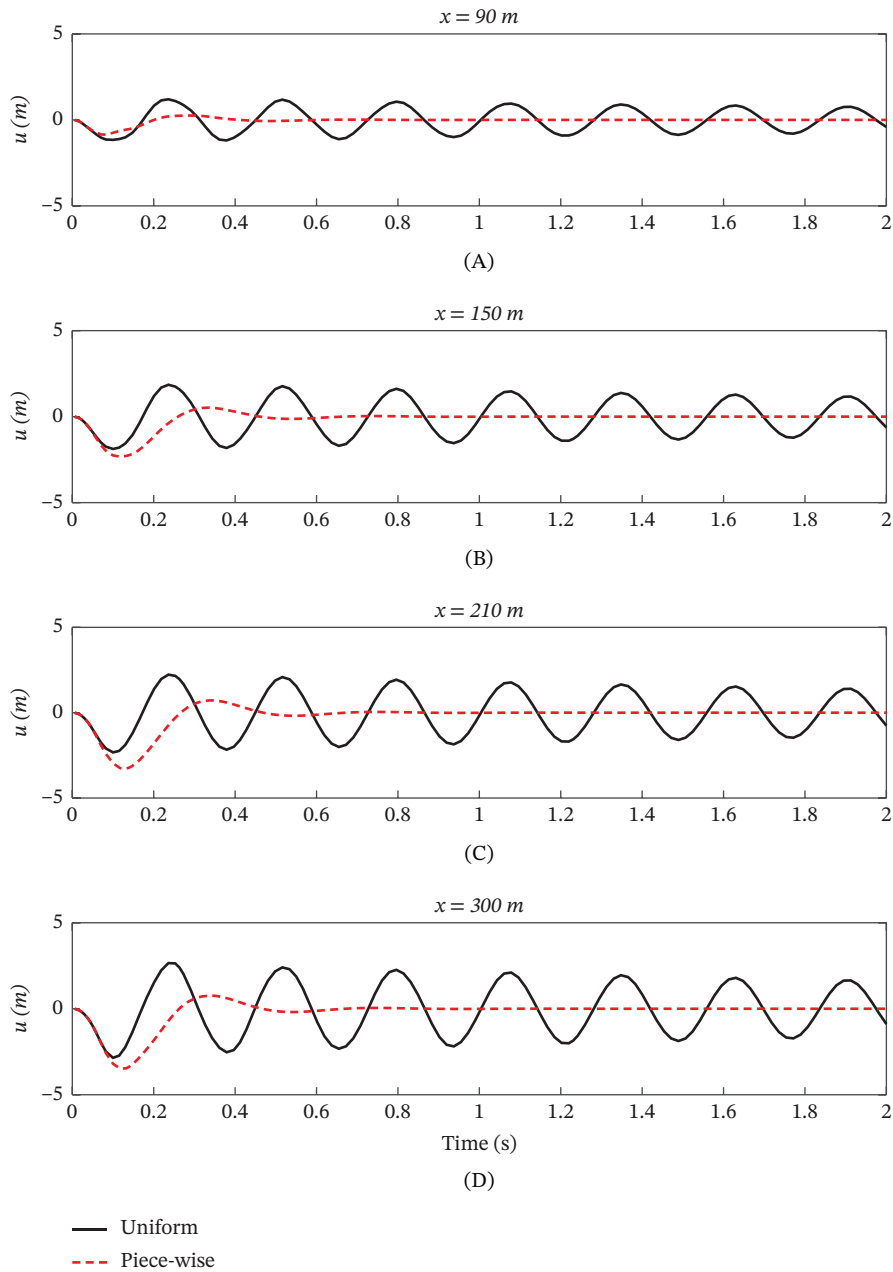


FIGURE 12 | Finite element results of axial dynamic responses of the piecewise space truss and uniform truss. (A) 90, (B) 150, (C) 210, and (D) 300 m.

In the finite element model, the total 909 Beam188 elements are used to simulate the piecewise space truss consisting of 153 battens and 150 longerons, and the total 900 Link180 elements are used to simulate the 300 diagonal cables. It should be noted that no initial cable force was applied to the six diagonal cables in the present finite element model. The diagonal cables were modeled as linear axial members, and only their stiffness and mass contributions were considered. This assumption was adopted to maintain consistency with the proposed analytical model, which focuses on axial vibration without prestress effects. The influence of cable pretension on the dynamic behavior of segmented space trusses is an important topic, but it is beyond the scope of the present study and will be considered in future work. When $k_1 = k_2 = 1e8/3 \text{ N/m}$ and $c_1 = c_2 = 1e8/3 \text{ N} \cdot \text{s/m}$, conduct modal analysis on the piecewise space truss using ANSYS, the results of their natural frequency comparison are shown in Table 4.

To assess agreement with the finite-element results, the accuracy of the natural frequencies of the analytical solution is calculated using the following expression:

$$e_i = \frac{|f_i^{AS} - f_i^{FEM}|}{f_i^{AS}} \times 100\%, \quad (25)$$

where f_i^{AS} is the i -th natural frequency of the analytical solution, and f_i^{FEM} is the i -th natural frequency of the finite element solution.

At the first eight natural frequencies, the relative errors of the analytical solution are no more than 1.3%, according to the error analysis. This shows that the piecewise model established in this study is highly accurate.

The first four mode shapes of the space piecewise truss are shown in Figures 8 and 9. The position of springs and dampers are marked, in which the displacement discontinuities are clearly seen, especially on the third mode shape in Figure 10. These visible displacement discontinuities at the segmentation points are consistent with the theoretical prediction of modal displacement jumps at compliant interfaces, indicating that interface compliance influences vibration transmission and contributes to localized energy dissipation. According to the previous dimensionless analysis, as the stiffness of the spring gradually increases, the piecewise space truss gradually becomes a uniform space truss, and the displacement discontinuities at the segmentation points gradually decrease until they disappear.

To further study the vibration characteristics of piecewise trusses, apply a global sinusoidal acceleration excitation to the piecewise space truss, acting on every mass element of the model. The magnitude of the sinusoidal acceleration excitation is $a(t) = a_0 \sin(2\pi ft)$ with amplitude $a_0 = 100 \text{ m/s}^2$ and frequency of $f = 40 \text{ Hz}$ along the axial direction. Measurement points were arranged at each segment of the space truss and at the free end, with a total of four measurement points located at $x = 90, x = 150, x = 210$, and $x = 300 \text{ m}$. Transient analysis simulation calculations were performed using ANSYS, and the time history curves of the axial displacement of the measurement points on the piecewise bars are shown in Figure 11. It clearly reveals that the noticeable vibration suppression effect achieved by the piecewise configuration with concentrated springs and dampers. The vibration amplitudes at critical locations of the piecewise space truss are markedly reduced compared with the uniform counterpart. This reduction is attributed to the displacement discontinuity induced by the concentrated springs, which disrupts the wave propagation and allows the concentrated dampers to effectively dissipate vibrational energy. However, the dynamic response in the piecewise truss shows that the dynamic response of the third piecewise bar is slightly bigger than uniform bar, that is, $x = 150 \text{ m}$. Dampers are installed in areas with small vibration displacement and velocity, where their energy absorption capacity is limited and energy dissipation efficiency is low, making them ineffective in suppressing the main vibration modes. Therefore, the vibration reduction effect depends on the stiffness, damping, and layout of the segmented interfaces, which should be considered carefully in practical design.

Furthermore, a condition was designed in which the global acceleration excitation was rectangular pulse excitation. The magnitude of the rectangular pulse acceleration excitation is $a = 898.2 \text{ m/s}^2$, and the excitation duration is 0.211 s . As shown in Figure 12, under impact excitation, the piecewise truss exhibits a much faster decay rate of vibrations, further confirming the enhanced damping capability provided by the localized dampers. These observations align well with the theoretical predictions derived from the piecewise bar model regarding the role of displacement discontinuity and localized energy dissipation in vibration mitigation, thereby validating the feasibility of piecewise space truss in actual LSTS applications.

4 | Conclusion

An analytical framework for the axial vibration of piecewise space truss incorporating concentrated spring–damper joints is

developed in this study. Derived via the PMPE and a dimensionless formulation, the framework yields characteristic equations for natural frequencies and discontinuous mode shapes, which are solved numerically and validated against finite-element results, with errors below 1.3%.

The results demonstrate that LSTS are intrinsically low-frequency flexible structures, and that segmentation further modifies their dynamic behavior relative to the corresponding uniform truss. Specifically, the compliant interfaces reduce the overall axial stiffness and therefore lower the natural frequencies, whereas the displacement discontinuities at the joints create localized compliance that enables energy storage and dissipation. With appropriately designed dampers, the segmented configuration reduces both steady-state and transient vibration responses and accelerates vibration decay under sinusoidal and rectangular pulse excitations. With the spring–damper configuration considered in this study, the segmented structure exhibits faster vibration decay under both sinusoidal and rectangular pulse excitations, demonstrating the potential of joint-level design for passive vibration control.

Beyond confirming the accuracy of the proposed model, this work establishes a theoretical foundation for the systematic design of vibration-suppressed large LSTS. The findings provide guidance for tailoring joint stiffness and damping in future deployable or modular aerospace structures, and open a pathway toward scalable, analytically tractable vibration control strategies for ultralarge space systems.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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